

CLASS 10 · CBSE · NCERT MATHEMATICS · SHORT NOTES

Chapter 8: Introduction to Trigonometry

Topic-wise short notes with solved examples

Trigonometry relates the sides of a right-angled triangle to its acute angles. This chapter covers the six ratios, their values for standard angles and the three basic identities.

1 Trigonometric ratios

- In a right-angled triangle ABC , right-angled at B , take the acute angle A . Then BC is the side **opposite** to A , AB is the side **adjacent** to A and AC is the **hypotenuse**.

Ratio	Definition	Ratio	Definition
$\sin A$	$\frac{\text{opposite}}{\text{hypotenuse}}$	$\operatorname{cosec} A$	$\frac{\text{hypotenuse}}{\text{opposite}}$
$\cos A$	$\frac{\text{adjacent}}{\text{hypotenuse}}$	$\sec A$	$\frac{\text{hypotenuse}}{\text{adjacent}}$
$\tan A$	$\frac{\text{opposite}}{\text{adjacent}}$	$\cot A$	$\frac{\text{adjacent}}{\text{opposite}}$

- Reciprocal relations:** $\operatorname{cosec} A = \frac{1}{\sin A}$, $\sec A = \frac{1}{\cos A}$, $\cot A = \frac{1}{\tan A}$.
- Quotient relations:** $\tan A = \frac{\sin A}{\cos A}$ and $\cot A = \frac{\cos A}{\sin A}$.
- The value of a ratio depends only on the angle, not on the size of the triangle.
- $\sin A$ is one symbol. It does not mean $\sin \times A$.

REMEMBER

The hypotenuse is the longest side, so $\sin A$ and $\cos A$ are never greater than 1, and $\sec A$ and $\operatorname{cosec} A$ are never less than 1.

2 Finding all ratios from one

- Step 1: draw a right-angled triangle and mark two sides using the given ratio, as k times the numbers.
- Step 2: find the third side with the Pythagoras theorem.
- Step 3: write the required ratios.

Example 2.1 If $\sin A = \frac{5}{13}$, find $\cos A$ and $\tan A$.

SOLUTION

Take opposite side = $5k$ and hypotenuse = $13k$.

Adjacent side = $\sqrt{(13k)^2 - (5k)^2} = \sqrt{144k^2} = 12k$

$$\cos A = \frac{12}{13} \text{ and } \tan A = \frac{5}{12}$$

Example 2.2 If $\tan \theta = \frac{3}{4}$, evaluate $\frac{1 - \tan^2 \theta}{1 + \tan^2 \theta}$ and show that it equals $\cos^2 \theta - \sin^2 \theta$.

SOLUTION

$$\frac{1 - \frac{9}{16}}{1 + \frac{9}{16}} = \frac{7/16}{25/16} = \frac{7}{25}$$

Opposite = $3k$, adjacent = $4k$, hypotenuse = $5k$, so $\sin \theta = \frac{3}{5}$ and $\cos \theta = \frac{4}{5}$

$$\cos^2 \theta - \sin^2 \theta = \frac{16}{25} - \frac{9}{25} = \frac{7}{25}$$

Both sides equal $\frac{7}{25}$.

3 Ratios of standard angles

Angle	0°	30°	45°	60°	90°
sin	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
cos	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
tan	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	Not defined
cosec	Not defined	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1
sec	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	Not defined
cot	Not defined	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0

- To remember the sin row: $\sqrt{\frac{0}{4}}, \sqrt{\frac{1}{4}}, \sqrt{\frac{2}{4}}, \sqrt{\frac{3}{4}}, \sqrt{\frac{4}{4}}$. The cos row is the sin row in reverse.
- As the angle increases from 0° to 90° , sin increases from 0 to 1 and cos decreases from 1 to 0.

Example 3.1 Evaluate $4 \sin^2 60^\circ + 3 \tan^2 30^\circ - 8 \sin 45^\circ \cos 45^\circ$.

SOLUTION

$$\begin{aligned} &= 4 \times \frac{3}{4} + 3 \times \frac{1}{3} - 8 \times \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} \\ &= 3 + 1 - 4 \\ &0 \end{aligned}$$

Example 3.2 If $\sin(A + B) = 1$ and $\cos(A - B) = \frac{\sqrt{3}}{2}$, where $0^\circ < A + B \leq 90^\circ$ and $A > B$, find A and B .

SOLUTION

$$\sin(A + B) = 1 = \sin 90^\circ, \text{ so } A + B = 90^\circ$$

$$\cos(A - B) = \frac{\sqrt{3}}{2} = \cos 30^\circ, \text{ so } A - B = 30^\circ$$

$$\text{Adding: } 2A = 120^\circ$$

$$A = 60^\circ \text{ and } B = 30^\circ$$

Example 3.3 In $\triangle ABC$, right-angled at B , $AC = 12$ cm and $\angle A = 60^\circ$. Find AB and BC .

SOLUTION

$$\cos 60^\circ = \frac{AB}{AC}, \text{ so } AB = 12 \times \frac{1}{2} = 6 \text{ cm}$$

$$\sin 60^\circ = \frac{BC}{AC}, \text{ so } BC = 12 \times \frac{\sqrt{3}}{2} = 6\sqrt{3} \text{ cm}$$

$$AB = 6 \text{ cm and } BC = 6\sqrt{3} \text{ cm}$$

REMEMBER

$\sin(A + B)$ is not $\sin A + \sin B$. Check with $A = B = 30^\circ$: $\sin 60^\circ = \frac{\sqrt{3}}{2}$, but $\sin 30^\circ + \sin 30^\circ = 1$.

4 Trigonometric identities

- An identity is true for every value of the angle for which both sides are defined.

$$\sin^2 A + \cos^2 A = 1 \quad 1 + \tan^2 A = \sec^2 A \quad 1 + \cot^2 A = \operatorname{cosec}^2 A$$

- Each identity can be rearranged: $\sin^2 A = 1 - \cos^2 A$, $\sec^2 A - \tan^2 A = 1$, $\operatorname{cosec}^2 A - \cot^2 A = 1$.
- $\sin^2 A$ means $(\sin A)^2$.

Example 4.1 Express $\cos A$ and $\tan A$ in terms of $\sin A$.

SOLUTION

From $\sin^2 A + \cos^2 A = 1$: $\cos A = \sqrt{1 - \sin^2 A}$

$$\tan A = \frac{\sin A}{\cos A} = \frac{\sin A}{\sqrt{1 - \sin^2 A}}$$

$$\cos A = \sqrt{1 - \sin^2 A} \text{ and } \tan A = \frac{\sin A}{\sqrt{1 - \sin^2 A}}$$

Example 4.2 Prove that $(1 + \tan^2 A)(1 - \sin A)(1 + \sin A) = 1$.

SOLUTION

$$\begin{aligned} \text{LHS} &= \sec^2 A (1 - \sin^2 A) \\ &= \sec^2 A \times \cos^2 A = \frac{1}{\cos^2 A} \times \cos^2 A \\ &= 1 = \text{RHS} \end{aligned}$$

Example 4.3 Prove that $\tan A + \cot A = \sec A \operatorname{cosec} A$.

SOLUTION

$$\begin{aligned} \text{LHS} &= \frac{\sin A}{\cos A} + \frac{\cos A}{\sin A} = \frac{\sin^2 A + \cos^2 A}{\sin A \cos A} \\ &= \frac{1}{\sin A \cos A} = \frac{1}{\cos A} \times \frac{1}{\sin A} \\ &= \sec A \operatorname{cosec} A = \text{RHS} \end{aligned}$$

Example 4.4 Prove that $\sqrt{\frac{1 - \cos A}{1 + \cos A}} = \operatorname{cosec} A - \cot A$.

SOLUTION

Multiply the numerator and denominator inside the root by $(1 - \cos A)$:

$$\begin{aligned} \text{LHS} &= \sqrt{\frac{(1 - \cos A)^2}{1 - \cos^2 A}} = \sqrt{\frac{(1 - \cos A)^2}{\sin^2 A}} = \frac{1 - \cos A}{\sin A} \\ &= \frac{1}{\sin A} - \frac{\cos A}{\sin A} \\ &= \operatorname{cosec} A - \cot A = \text{RHS} \end{aligned}$$

5 How to prove an identity

- Start from the more complicated side and work towards the simpler one.
- If nothing else suggests itself, change every ratio to $\sin$ and $\cos$.
- For a sum of fractions, take the LCM. For $1 \pm \sin A$ or $1 \pm \cos A$ in a denominator, multiply by the conjugate.
- Look for $1 - \sin^2 A$, $1 - \cos^2 A$, $\sec^2 A - 1$ and $\operatorname{cosec}^2 A - 1$: each one is a single square.
- Never move terms from one side to the other in a proof. Work on one side only, or simplify both sides separately.

6 Common mistakes

- Writing $\sin A = \frac{\text{opposite}}{\text{adjacent}}$. That is $\tan A$.
- Treating $\sin^2 A$ as $\sin A^2$ or $\sin 2A$.
- Using $\tan 90^\circ$ or $\sec 90^\circ$ as a number. They are not defined.
- Cancelling $\sin$ from $\frac{\sin A}{\sin B}$ to get $\frac{A}{B}$.